



MongoDB Log Anomaly & Security Monitor

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Project Details

Project Overview

AI-powered real-time monitoring platform for MongoDB logs that automatically detects:

- Security threats
- Anomalies
- Slow queries
- Performance issues
- Suspicious activities

Users

- ▶ **Admin:** Full system control — manage users/roles, settings, model lifecycle, and all data (create/update/delete).
- ▶ **Analyst (User):** Investigate and triage alerts, annotate/resolve anomalies, run searches, and request model retraining.
- ▶ **Viewer:** Read-only access to dashboards, logs, and alert summaries without modification rights.

Core Features

- Real-time log monitoring
- AI anomaly detection
- NLP-based log classification
- Interactive dashboard
- Live alerts & notifications
- Role-based access control
- Docker & Kubernetes deployment

Purpose: Centralized log ingestion, AI-driven anomaly detection, classification, alerting, and dashboarding for security/operational monitoring.

Scope: Ingest logs, store in MongoDB, run ML/NLP models to detect anomalies/classify logs, surface alerts to users via backend APIs and frontend UI, and support notifications and dataset management.

TECHNOLOGY STACK

FRONTEND TECHNOLOGIES



React.js — Used for building the frontend user interface and dashboards



Vite — Used as the frontend development and build tool



Tailwind CSS — Used for responsive UI styling and modern design



Socket.IO Client — Used for real-time logs, alerts, and anomaly updates

DATABASE TECHNOLOGIES



MongoDB — Main database used for storing logs, anomalies, alerts, reports, and users



Mongoose — Used for MongoDB schema and model management

DEVOPS & DEPLOYMENT TECHNOLOGIES



Docker — Used for containerizing frontend, backend, and AI services



Kubernetes — Used for orchestration, scaling, and deployment



NGINX — Used as reverse proxy and load balancer



Apache Kafka — Used for real-time log streaming and message processing



RabbitMQ — Not used in the current implementation

BACKEND TECHNOLOGIES



Node.js — Backend runtime environment



Express.js — Used for building REST APIs and backend services



JWT Authentication — Used for secure authentication and authorization



REST APIs — Used for communication between frontend, backend, and AI service



Socket.IO Server — Used for real-time communication and live monitoring

AI / MACHINE LEARNING TECHNOLOGIES



Python — Used for AI and machine learning services



FastAPI — Used to create AI microservice APIs



scikit-learn — Used for anomaly detection models



spaCy — Used for NLP-based log classification



NumPy — Used for numerical and machine learning operations



NLTK — Not used in the current implementation

How & Where MongoDB and AI are Used?

MongoDB Usage

- ▶ MongoDB is the main database used in the project
- ▶ Stores:
 - Raw MongoDB logs
 - Processed log data
 - Detected anomalies
 - Alerts & notifications
 - User details and settings
 - SIEM training datasets
- ▶ MongoDB helps with:
 - Real-time log storage
 - Fast data retrieval
 - Historical log analysis
 - Scalable high-volume data handling

AI Usage

- ▶ AI service is built using Python + FastAPI
- ▶ AI analyzes MongoDB logs in real time

AI Components

- ▶ `anomaly_detector.py`
 - Detects unusual activities and anomalies
 - Uses Isolation Forest & LOF algorithms
- ▶ `log_classifier.py` : Uses NLP to classify logs
 - Identifies:
 - Security threats
 - Slow queries
 - Authentication failures
 - Replication errors
- ▶ `log_processor.py`
 - Parses and converts raw logs into AI-ready features

HOW AND WHERE MONGODB & AI ARE USED IN THIS PROJECT

Real-time Log Anomaly Detection & Security Monitoring System

1. LOG SOURCES

Logs come from different sources



MongoDB Logs



Log Files



External Systems



Log Simulator

2. INGESTION (BACKEND)

Logs are received by Backend API

POST /api/logs

200 OK

```
{
  "timestamp": "2024-11-20T12:30:45Z",
  "level": "ERROR",
  "component": "QUERY",
  "message": "Slow query detected",
  "duration": 5324,
  "context": { "collection": "users" }
}
```



Express

3. MONGODB (CENTRAL DATABASE)

Stores all logs, results, alerts, settings and datasets



Logs Collection
Raw and processed logs

Settings Collection
System settings & thresholds

Anomalies Collection
AI detected anomalies

Dataset Collection
Labeled data for training

Alerts Collection
Critical alerts & status

Users Collection
User info & roles

KAFKA (STREAMING LAYER)

Decouples services for real-time streaming



raw-logs topic

processed-logs topic

anomalies topic

5. DASHBOARD (FRONTEND)

Real-time monitoring and visualization



6. ALERTS & NOTIFICATIONS

Critical alerts are sent in real-time

New Critical Alert

Slow Query Detected

Type: performance
Severity: high
Score: 0.87
Time: 2024-11-20 12:30:45

[View Details](#) [Acknowledge](#)

Email

Slack

WebSocket (Real-time)

SMS / Other

4. AI SERVICE (PYTHON)

AI analyzes logs for anomalies and classification



1. Log Ingestion
Reads logs from MongoDB / Kafka

```
INFO Ingested log: {"timestamp": "2024-11-20T12:30:45Z",
"level": "ERROR", "component": "QUERY", ... }
INFO Ingested log: {"timestamp": "2024-11-20T12:30:45Z",
"level": "INFO", "component": "QUERY", ... }
```



2. Log Processing
Parses and converts logs to structured features

```
{
  "timestamp": "2024-11-20T12:30:45Z",
  "level": "ERROR",
  "component": "QUERY",
  "duration": 5324,
  "collection": "users"
}
```



3. AI Anomaly Detection
Detects anomalies using ML models (Isolation Forest, LOF etc.)

Anomaly Score: 0.87
Anomaly Type: performance
Algorithm: isolation_forest
Severity: high



4. Log Classification (NLP)
Classifies log type using NLP (spaCy)

Classification: slow_query
Confidence: 0.92
Entities: ["collection", "users"]

AI results are saved back to MongoDB

HOW MONGODB IS USED

- ✓ Stores all raw, processed logs and AI results
- ✓ Handles high-volume log ingestion
- ✓ Provides fast queries for analysis and dashboards
- ✓ Stores anomalies, alerts and user feedback
- ✓ Stores training datasets for AI model improvement

HOW AI IS USED

- ✓ Detects anomalies using ML (Isolation Forest, LOF)
- ✓ Classifies logs using NLP (spaCy)
- ✓ Extracts patterns and meaning from logs
- ✓ Assigns anomaly score, severity and type
- ✓ Continuously improves using feedback and dataset

TECH STACK



MongoDB



Express.js



React



Node.js



Python



FastAPI



scikit-learn



spaCy



Kafka



Docker



Kubernetes

Problem Statement & Solution

Problem Statement:

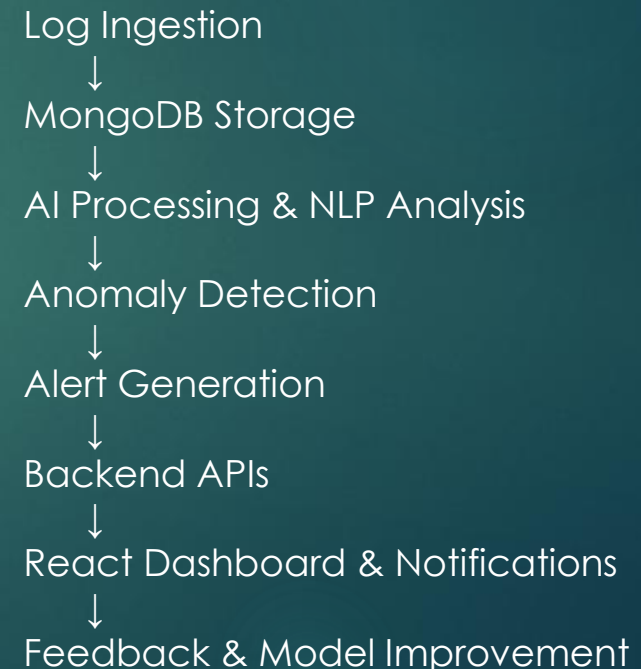
- MongoDB generates huge volumes of logs continuously, making manual monitoring difficult
- Logs are noisy, complex, and contain mixed information from security, performance, and operational events
- Detecting real security threats and meaningful anomalies with low false-positive rates is challenging
- Critical incidents such as unauthorized access, slow queries, and replication failures may go unnoticed
- Operators need fast alerting, contextualized monitoring, and an auditable incident history
- AI/ML models require quality labeled datasets and historical logs for accurate detection and retraining

Proposed Solution:

MongoDB Log Anomaly & Security Monitor

- Collects MongoDB logs in real time through ingestion pipelines
- Uses AI & NLP models to preprocess, classify, and analyze logs
- Detects anomalies, suspicious activities, and performance issues automatically
- Stores raw logs, processed data, anomalies, and alerts in MongoDB
- Provides real-time dashboards and live alerts using React and Socket.IO
- Supports notifications through Email, Slack, Telegram, and in-app alerts
- Uses feedback and labeled datasets stored in MongoDB to continuously improve AI models

System Flow



How MongoDB Helps the AI

MongoDB is the data backbone that powers smarter, faster and more accurate AI.



Scalable Storage for High-Ingest Rate

High-throughput writes for raw logs enable durable capture of events without blocking inference.



Flexible Document Model

Stores nested parsed fields, variable schemas, and metadata needed for ML features.



Fast Retrieval & Indexing

Indexes on timestamp, host, or labels support efficient queries for model training, feature extraction, and investigation.



Dataset Management

Store labeled examples and versioned dataset records to support supervised retraining and evaluation.



Real-time Triggers

Change streams (or polling) enable near-real-time notification of new logs to the AI pipeline for low-latency inference.



Auditability & Provenance

Raw + processed documents kept together so predictions are explainable and reproducible.



MongoDB



Central Data Hub



Raw Logs

Stores high volume raw logs in real time



Processed Data

Parsed, normalized and enriched logs



AI Features

Extracted features used for ML models



Anomalies & Alerts

AI results, scores, and alert history



Labeled Datasets

Labeled examples for training & evaluation



Metadata & Settings

AI model configs, thresholds and system settings



AI / ML Pipeline



Data Ingestion

New logs captured and stored in MongoDB



Data Retrieval

AI queries MongoDB for training & inference



Model Training

Uses historical & labeled data from MongoDB



Inference / Detection

AI detects anomalies and classifies logs



Results Stored Back

Predictions & feedback saved in MongoDB

Key Benefits Delivered



Reduces operator noise via ML scoring and classification.



Enables automated alert generation with contextual evidence.



Provides persistent labeled data to improve models over time.



Supports scalable deployment using Docker/Kubernetes and MongoDB's scaling.



MongoDB Logs



Kafka Streaming



Python AI Service



MongoDB Storage



Dashboard & Alerts



User Feedback



Future Enhancements

Implement Advanced AI Models:

- ✓ LSTM
- ✓ Transformers
- ✓ Deep Learning models

Integrate MongoDB Atlas Vector Search for:

- ✓ Similarity detection
- ✓ Root cause analysis

Add Explainable AI Features:

- ✓ Feature importance
- ✓ Prediction explanations
- ✓ AI confidence analysis

Extend Multi-Database Support:

- ✓ MySQL
- ✓ PostgreSQL
- ✓ Redis

Add Cloud Platform Integrations:

- ✓ AWS CloudWatch
- ✓ Azure Monitor
- ✓ Google Cloud Logging

Implement AI Feedback Learning:

- ✓ Users can label alerts
- ✓ Continuous AI model improvement
- ✓ Better anomaly detection accuracy

Improve Scalability & Performance:

- ✓ Kubernetes auto-scaling
- ✓ Distributed Kafka streaming
- ✓ MongoDB sharding

Build Advanced Analytics Dashboard:

- ✓ Predictive insights
- ✓ Threat trend analysis
- ✓ Security heatmaps



THANK YOU